

AdaBoost from Scratch in Python

Contents

1. Overview	2
2. The data	3
3. Stage 1: weights, stump, error and alpha	3
4. Updating the weights	5
5. Why the update uses the exponential	5
6. Normalising and drawing the next dataset	6
7. Stage 2, and a stump with no mistakes	7
8. Stage 3	8
9. Prediction: the weighted vote	8
10. Weights instead of upsampling: how scikit-learn does it	9
11. Summary	11
12. Key terms	11

Where this fits: Pipeline map step 8 (Model). See the [Course map](#).



- **Builds on:** Decision trees ([Note 100](#)); Boosting ([Note 101](#)); Hyperparameter tuning ([Note 111](#)).

1. Overview

Key point: We code three stages of AdaBoost by hand on 10 rows: weights, stump, error, alpha, weight update, normalisation and upsampling, then the weighted vote. The result matches the steps of the step-by-step Note.

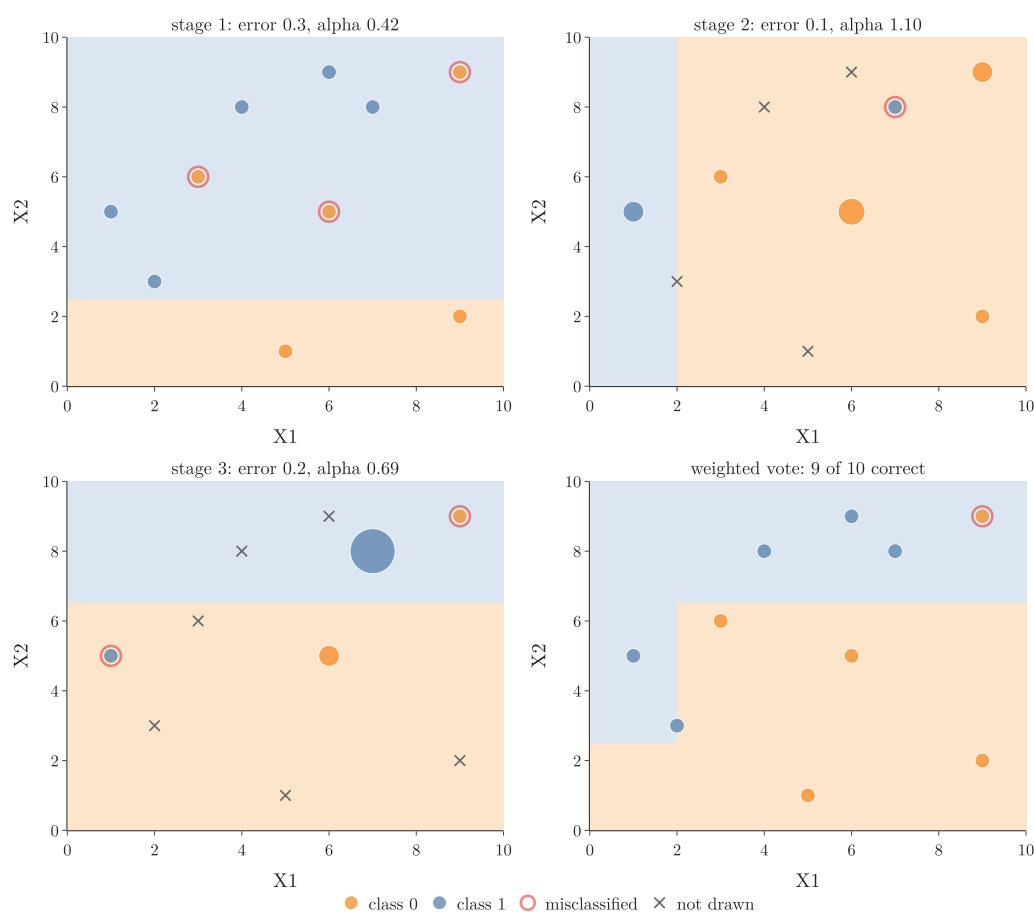


Figure 1: Three stages of AdaBoost on the 10-row toy data, and their weighted vote. Marker size shows how many copies of a row are in that stage's dataset; crosses are rows not drawn

The [AdaBoost step-by-step Note](#) went through one stage on paper. Here we write each step in Python with pandas, NumPy and scikit-learn, and run three stages (Figure 1). The

Notebook (`notebook.ipynb`) holds the full code; `images/figs.py` draws the figures with the same data and random seed.

Along the way we see two things the paper version did not show: why the weight update uses the exponential, and what to do when a stump makes no mistakes at all.

2. The data

Key point: 10 rows, inputs X1 and X2, classes 1 and 0; no single straight cut separates them.

Row	X1	X2	label
0	1	5	1
1	2	3	1
2	3	6	0
3	4	8	1
4	5	1	0
5	6	9	1
6	6	5	0
7	7	8	1
8	9	9	0
9	9	2	0

Rows are numbered from 0, as pandas does. The classes are mixed (Figure 1, top left), so the data is non-linear: a stump alone will make mistakes, which is exactly what we want to watch.

The labels are stored as 1 and 0, the way scikit-learn expects them. For the final vote we turn them into +1 and -1 (section 9).

Python: Building the table.

```
import pandas as pd
```

```
df = pd.DataFrame()
```

```
df["X1"] = [1, 2, 3, 4, 5, 6, 6, 7, 9, 9]
```

```
df["X2"] = [5, 3, 6, 8, 1, 9, 5, 8, 9, 2]
```

```
df["label"] = [1, 1, 0, 1, 0, 1, 0, 1, 0, 0]
```

Assigning a list to `df["X1"]` creates a column named X1.

3. Stage 1: weights, stump, error and alpha

Key point: Every row gets 0.1; the first stump is “X2 at most 2.5 means class 0”; it misclassifies rows 2, 6 and 8, so error = 0.3 and alpha = 0.42.

Weights. With 10 rows, each row starts at $1/10 = 0.1$.

The stump. A `DecisionTreeClassifier` with `max_depth=1` is a decision stump. Trained on all 10 rows, it chooses $X2 \leq 2.5$: below, class 0; above, class 1 (Figure 1, top left).

Python: Weights and the first stump.

```
from sklearn.tree import DecisionTreeClassifier

df["weights"] = 1 / df.shape[0]    # 0.1 for every row
X = df[["X1", "X2"]].values
y = df["label"].values
dt1 = DecisionTreeClassifier(max_depth=1, random_state=0)
dt1.fit(X, y)
df["y_pred"] = dt1.predict(X)
df.shape[0] is the number of rows. export_text(dt1) from sklearn.tree prints the
split as text.
```

Extra: On this data the cut “ $X1 \leq 2.5$ ” is exactly as good as “ $X2 \leq 2.5$ ”: both get 3 rows wrong. When two cuts tie, scikit-learn breaks the tie by shuffling the columns at random, so the chosen cut depends on `random_state`. With `random_state=0` we get the X2 cut; other values give the X1 cut.

Mistakes. Comparing `label` with `y_pred` row by row, the stump is wrong on rows **2, 6 and 8**: three class-0 rows in the class-1 region.

Error and alpha. The error is the total weight of those rows, $0.1 + 0.1 + 0.1 = 0.3$. Using the formula of the step-by-step Note, section 6:

$$\alpha_1 = \frac{1}{2} \ln \left(\frac{1 - 0.3}{0.3} \right) = \frac{1}{2} \ln(2.333) = \frac{1}{2} \times 0.847 = 0.4236$$

Python: Error and alpha as functions.

```
import numpy as np

def model_error(d):
    wrong = d["label"] != d["y_pred"]
    return d.loc[wrong, "weights"].sum()

def calculate_model_weight(error, eps=1e-10):
    return 0.5 * np.log((1 - error + eps) / (error + eps))

alpha1 = calculate_model_weight(model_error(df))    # 0.4236
d.loc[wrong, "weights"] keeps only the rows where wrong is True. np.log is the
natural log. eps is explained in section 7.
```

4. Updating the weights

Key point: The three mistakes go from 0.1 to 0.1528; the seven correct rows from 0.1 to 0.0655.

With $\alpha_1 = 0.4236$ (the step-by-step Note, section 7):

$$\text{misclassified: } 0.1 \times e^{0.4236} = 0.1 \times 1.5275 = 0.1528$$

$$\text{correct: } 0.1 \times e^{-0.4236} = 0.1 \times 0.6547 = 0.0655$$

Python: The update for every row at once.

```
def update_row_weights(d, alpha):
    correct = d["label"] == d["y_pred"]
    return np.where(correct,
                    d["weights"] * np.exp(-alpha),
                    d["weights"] * np.exp(alpha))

df["updated_weights"] = update_row_weights(df, alpha1)
np.where(condition, a, b) takes a where the condition is True and b where it is
False, row by row.
```

5. Why the update uses the exponential

Key point: e^α grows fast with alpha: a stump we trust a lot changes the weights a lot; a stump we barely trust changes them little.

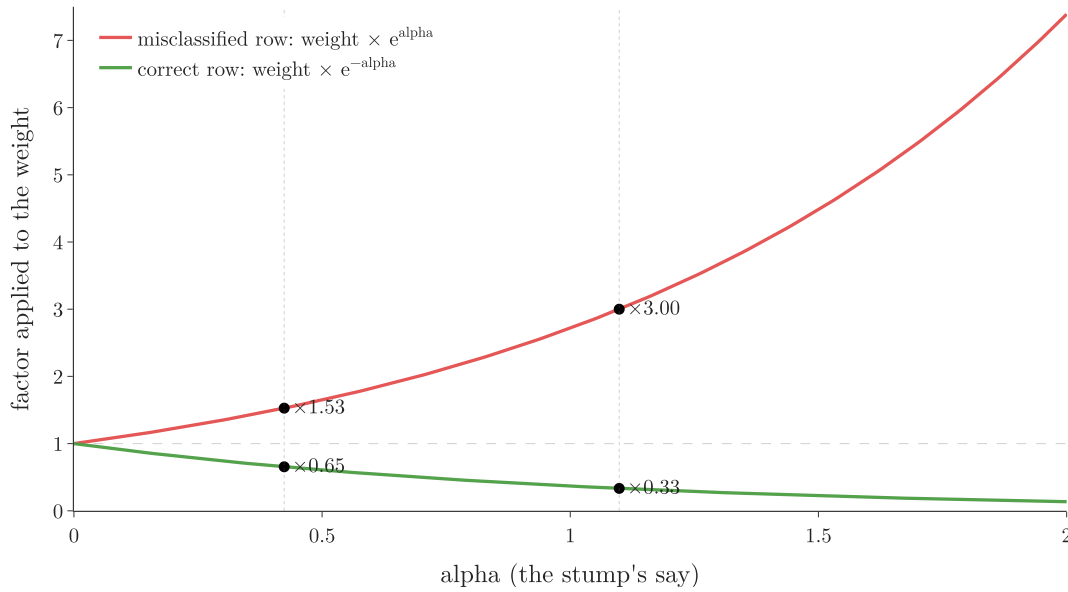


Figure 2: The factors e^α (mistakes) and $e^{-\alpha}$ (correct rows) against alpha, marked at our stage 1 and stage 2 alphas

Figure 2 plots the two factors.

- e^α (**red**) is 1 at $\alpha = 0$ and rises quickly. A large alpha means a trustworthy stump, one with few mistakes. If such a stump still gets a row wrong, that row is genuinely hard, and its weight is multiplied by a big number: $\times 3.00$ for our stage 2 stump ($\alpha = 1.10$).
- $e^{-\alpha}$ (**green**) is the mirror image: below 1 for any positive alpha, and smaller the larger alpha is. Rows a trustworthy stump gets right lose most of their weight: $\times 0.33$ at $\alpha = 1.10$.
- A stump we barely trust (small alpha) changes the weights only a little: $\times 1.53$ and $\times 0.65$ for our stage 1 stump ($\alpha = 0.42$).

So the size of the update follows how much we trust the stump that made it.

6. Normalising and drawing the next dataset

Key point: The updated weights add to 0.9165; after dividing, mistakes weigh 0.1667 and correct rows 0.0714. Ten random draws then give the dataset for stage 2.

Normalise. The updated weights add up to $3 \times 0.1528 + 7 \times 0.0655 = 0.9165$. Dividing each by 0.9165:

$$\text{misclassified: } \frac{0.1528}{0.9165} = 0.1667 \quad \text{correct: } \frac{0.0655}{0.9165} = 0.0714$$

Ranges. The running total of the normalised weights, `np.cumsum`, gives each row's upper end; its lower end is the upper end minus its own weight. Row 0 owns 0 to 0.0714, row 1

owns 0.0714 to 0.1429, row 2 owns 0.1429 to 0.3095, and so on.

Upsampling. We draw 10 random numbers between 0 and 1 and, for each, pick the row whose range it falls in (the step-by-step Note, section 9).

Python: Ranges and the new dataset.

```
df["normalized_weights"] = (df["updated_weights"]
                             / df["updated_weights"].sum())
df["cumsum_upper"] = np.cumsum(df["normalized_weights"])
```

```
rng = np.random.default_rng(0)
```

```
def create_new_dataset(d):
    indices = []
    for _ in range(d.shape[0]):
        a = rng.random()
        # first row whose upper end is above a
        upper = d["cumsum_upper"].values
        indices.append(int(np.argmax(upper > a)))
    return indices
```

```
index_values = create_new_dataset(df)
second_df = (df.iloc[index_values][["X1", "X2", "label"]]
             .reset_index(names="row"))
```

`rng.random()` returns a random number between 0 and 1; seed 0 makes the draw repeatable. `upper > a` is a list of True/False; `np.argmax` returns the position of the first True. `reset_index(names="row")` numbers the new rows 0 to 9 and keeps the original row number in a column `row`.

Our draw picks rows 6, 2, 0, 0, 8, 8, 6, 7, 6, 9. The three mistakes (rows 2, 6 and 8) take 6 of the 10 places; rows 1, 3, 4 and 5 are not drawn at all (Figure 1, top right: crosses).

Weights start again. In the new dataset every row gets weight 0.1 again. The upsampling has already done the reweighting: a row drawn three times counts three times.

Extra: `np.random.default_rng(0)` is NumPy's current way to make random numbers. Older code calls `np.random.random()` with no seed, so every run draws different rows and gives different stumps. Fixing the seed makes the results repeatable. NumPy can also draw by weight in one line: `rng.choice(10, size=10, p=weights)`.

7. Stage 2, and a stump with no mistakes

Key point: The second stump is "X1 at most 2 means class 1"; it gets one row wrong, so error = 0.1 and alpha = 1.10, a bigger say than the first stump's.

On `second_df` we repeat every step:

1. weights 0.1 each;

2. the new stump is $\mathbf{X1} \leq 2$: left, class 1; right, class 0 (Figure 1, top right);
3. it gets one row wrong, original row 7 (class 1 at $X1 = 7$), so the error is 0.1;
4. $\alpha_2 = 0.5 \times \ln(0.9/0.1) = 0.5 \times \ln 9 = 1.0986$.

The second stump has a larger say than the first (1.10 against 0.42) because it made fewer mistakes.

The update follows (Figure 2): the mistake becomes $0.1 \times e^{1.0986} = 0.3$, the nine correct rows $0.1 \times e^{-1.0986} = 0.0333$. Their sum is $0.3 + 9 \times 0.0333 = 0.6$; after normalising, the mistake weighs **0.5** and every other row 0.0556. The draw for stage 3 gives row 7 seven times out of ten.

When a stump makes no mistakes. On a small dataset a stump can classify every row correctly. Its error is then 0, and the formula breaks:

$$\alpha = \frac{1}{2} \ln \left(\frac{1-0}{0} \right) \quad \text{divides by zero}$$

The fix is to add a tiny number, such as 10^{-10} , to the error before dividing; that is the **eps** in `calculate_model_weight`. Such a stump then gets a huge but finite alpha. `scikit-learn` handles this case itself: it stops boosting early, because a perfect stump leaves nothing to fix.

8. Stage 3

Key point: The third stump, trained on the third dataset, is “X2 at most 6.5 means class 0”; error 0.2, alpha 0.69.

The third dataset is rows 7, 6, 7, 6, 7, 0, 7, 7, 8, 7. The new stump is $\mathbf{X2} \leq 6.5$: below, class 0; above, class 1 (Figure 1, bottom left). It gets two rows wrong, original rows 0 and 8, so the error is 0.2 and

$$\alpha_3 = \frac{1}{2} \ln \left(\frac{1-0.2}{0.2} \right) = \frac{1}{2} \ln 4 = 0.6931$$

Extra: A common slip when copying the stage code is to train the third stump on the second dataset while comparing its predictions with the labels of the third. The predictions then belong to different rows from the labels, the “error” comes out meaninglessly high (0.7, say), and alpha turns negative. Each stump must be trained and scored on the same dataset; the Notebook keeps one table per stage for that reason.

9. Prediction: the weighted vote

Key point: Convert each stump’s 1/0 to +1/-1, weight by alpha, add, take the sign. Row (1, 5) gets +0.83: class 1. Row (9, 9) gets +0.02: class 1, which is wrong.

We have $\alpha_1 = 0.4236$, $\alpha_2 = 1.0986$, $\alpha_3 = 0.6931$. A stump predicts 1 or 0; for the vote (the [AdaBoost intuition Note](#), section 5) we turn 1 into +1 and 0 into -1 with $2p - 1$.

Query (1, 5), true class 1. The stumps say: $X_2 = 5 > 2.5$, so +1; $X_1 = 1 \leq 2$, so +1; $X_2 = 5 \leq 6.5$, so -1.

$$0.4236 \times (+1) + 1.0986 \times (+1) + 0.6931 \times (-1) = 0.8291$$

Positive, so +1: class 1, **correct**.

Query (9, 9), true class 0. The stumps say: $X_2 = 9 > 2.5$, so +1; $X_1 = 9 > 2$, so -1; $X_2 = 9 > 6.5$, so +1.

$$0.4236 \times (+1) + 1.0986 \times (-1) + 0.6931 \times (+1) = 0.0181$$

Positive, so +1: class 1, **wrong**, though only just. On all 10 training rows the three stumps get 9 right; row 8 is the one they miss (Figure 1, bottom right). With more stages, later stumps would keep working on it.

Python: The weighted vote.

```
def to_pm(p):          # 1 -> +1, 0 -> -1
    return 2 * p - 1
```

```
def adaboost_predict(query):
    query = np.array(query).reshape(1, 2)
    votes = [to_pm(m.predict(query)[0])
              for m in (dt1, dt2, dt3)]
    total = (alpha1 * votes[0] + alpha2 * votes[1]
              + alpha3 * votes[2])
    return np.sign(total)
```

```
adaboost_predict([1, 5])    # 1.0
```

`reshape(1, 2)` turns the pair into a table of one row and two columns, the shape `predict` expects. `np.sign` returns 1.0, -1.0 or 0.0.

10. Weights instead of upsampling: how scikit-learn does it

Key point: scikit-learn hands the weights to the stump with `sample_weight` instead of drawing rows, and its `alpha` has no factor $1/2$; the decisions are the same. With the same tie-breaking, our loop reproduces `AdaBoostClassifier` exactly.

Upsampling is random: another seed draws other rows and can give other stumps. scikit-learn's `AdaBoostClassifier` avoids the randomness. It never draws rows; it trains each stump with `fit(X, y, sample_weight=w)`, so the tree counts every row in proportion to its weight when it compares cuts.

It also uses a slightly different but equivalent bookkeeping, the **SAMME** algorithm:

- $\alpha = \ln((1 - \text{error})/\text{error})$, twice our α (no factor 0.5);
- only the mistakes are multiplied, by e^α ; correct rows keep their weight; then all are normalised.

Both changes leave the result unchanged. Doubling every alpha doubles the vote total but never changes its sign. And multiplying the mistakes by $e^{2\alpha}$ gives the same normalised weights as multiplying the mistakes by e^α and the correct rows by $e^{-\alpha}$: only the ratio between the two groups matters.

Python: AdaBoost with sample weights, checked against scikit-learn.

```
from sklearn.ensemble import AdaBoostClassifier
```

```
def adaboost_weights(X, y, seeds):
    w = np.full(len(y), 1 / len(y))
    stumps, alphas = [], []
    for seed in seeds:
        stump = DecisionTreeClassifier(max_depth=1,
                                       random_state=seed)
        stump.fit(X, y, sample_weight=w)
        wrong = stump.predict(X) != y
        err = w[wrong].sum()
        alpha = np.log((1 - err) / err)    # SAMME
        w = w * np.exp(alpha * wrong)     # mistakes only
        w = w / w.sum()
        stumps.append(stump)
        alphas.append(alpha)
    return stumps, np.array(alphas)
```

```
abc = AdaBoostClassifier(DecisionTreeClassifier(max_depth=1),
                        n_estimators=3, random_state=0)
```

```
abc.fit(X, y)
```

```
seeds = [s.random_state for s in abc.estimators_]
```

```
stumps, alphas = adaboost_weights(X, y, seeds)
```

$\alpha * \text{wrong}$ is α for a mistake (True counts as 1) and 0 for a correct row ($e^0 = 1$). scikit-learn gives each stump its own `random_state`; reusing them makes ties break the same way.

Both give alphas 0.8473, 1.2993 and 1.8458 (`abc.estimator_weights_`) and the same predictions. Without random draws, the three weighted stumps classify **all 10 rows** correctly. The first alpha, 0.8473, is exactly twice our hand-computed 0.4236.

11. Summary

Stage	Data	Stump	Mistakes	Error	Alpha
1	all 10 rows	$X_2 \leq 2.5$: class 0	rows 2, 6, 8	0.3	0.4236
2	drawn: 6, 2, 0, 0, 8, 8, 6, 7, 6, 9	$X_1 \leq 2$: class 1	row 7	0.1	1.0986
3	drawn: 7, 6, 7, 6, 7, 0, 7, 7, 8, 7	$X_2 \leq 6.5$: class 0	rows 0, 8	0.2	0.6931

- Every step of a stage is a few lines of pandas and NumPy: weights, stump, error, alpha, update, normalise, draw.
- The exponential update makes trusted stumps change the weights a lot and doubtful ones a little.
- An error of 0 breaks the alpha formula; a tiny `eps` keeps it finite.
- Each stump must be trained and scored on its own stage's dataset.
- Our three upsampled stumps get 9 of 10 training rows right; scikit-learn's weighted stumps (SAMME, `sample_weight`) get all 10.

12. Key terms

Term	Meaning
<code>sample_weight</code>	The argument of <code>fit</code> that tells a scikit-learn model how much each row counts
SAMME	The AdaBoost variant in scikit-learn: alpha without the factor 1/2, only misclassified rows reweighted; same decisions
Random seed	A number that fixes a random number generator so that a run can be repeated exactly